



FIRST SEMESTER 2026-27 COURSE

HANDOUT PART II

Date: 29.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course Number : PHA G 534
Course Title : Separation and Structure Elucidation Techniques
Instructor-in-Charge : Prof. Paul Atish Tulshiram
Instructor(s) : NA
Tutorial/Practical Instructors : NA

- 1. Course Description:** Advance separation methods for complex Natural, Semi-synthetic and Synthetic New Chemical Entities (NCEs) using LC-MS, LC-MS-MS, LC-NMR, GC-MS, GC-MS-MS, CE-MS, Supercritical Fluid Chromatography. Structure elucidation of NCEs using 2D, 3D (^1H and ^{13}C) NMR spectroscopy and X-ray diffraction technique. Prediction of structures using Software tools.
- 2. Scope and Objective:** a. This course deals with the study of advanced techniques for separation and structure elucidation of organic molecules, especially of pharmaceutical importance.
- 3. Course learning outcomes:** Upon completion of the course, the students will be able to use all the techniques of the course for in-depth study of any pharmaceutical drugs and their products.
- 4. Text Books and references:**

TB 1.Handbook of Advanced Chromatography /Mass Spectrometry Techniques by Michal Holcapek
WM. Craig Byrdwell, 1st Edn, Academic Press and AOCS Press, 2017

TB 2.Introduction to Spectroscopy by Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan, 5th Edn, Cengage Learning India Private Limited, 2015

RB1.On-line LC-NMR and Related Techniques by Klaus Albert, 2002, John Wiley & Sons, Ltd. England.

RB2.Hyphenated and Alternative Methods of Detection in Chromatography by R. Andrew Shalliker, 2019, CRC Press, UK.

RB3. Capillary Electrophoresis - Mass Spectrometry (CE-MS): Principles and Applications by Gerhardus de Jong, 2016, John Wiley & Sons, Ltd. England.

RB4. Modern Supercritical Fluid Chromatography by Larry M. Miller, J. David Pinkston, and Larry T. Taylor, 2019, John Wiley & Sons, Inc. USA

Journal Article 1: E. S. Ameh. A review of basic crystallography and x-ray diffraction applications. *Int J Adv Manuf Technol* (2019) 105, pages 3289–3302.



5. Method of conduct of the course: The Course will be taught in a classroom with the help of PowerPoint presentation videos and a blackboard. It will involve discussions on case studies. Students will be invited to present their point on the topic of discussion. It will also have demonstrations of high-end instruments.

6. Course Plan:

Week no.	Module	Lecture no.	Reference	Learning Outcome
1	I [LC-MS]	[1-3]	TB-1 RB-3	Students will learn the basics of LC-MS,
2	I [LC-MS- MS]	[4-6]	TB-1 RB-3	Students will learn the basics of LC-MS/MS, and its applications in drug discovery and development
3	II [1D NMR]	[7-9]	TB-2	Students will learn about the structure elucidations of small molecules using 1D NMR experiments
4	II [2D NMR]	[10-12]	TB-2 RB-1	Students will learn about the structure elucidations of small molecules using 2D NMR experiments
5	II [2D NMR]	[13-15]	TB-2 RB-1	Students will learn about the structure elucidations of small molecules using 2D NMR experiments
6	II [3D NMR]	[16-18]	RB-1	Students will learn about the structure elucidations of large molecules using 3D NMR experiments
7	II [LC-NMR]	[19-21]	TB-1 RB-1	Students will learn the basics of LC-NMR
8	III [GC-MS]	[22-24]	RB-2	Students will learn the basics and method development of GC-MS
9	III [GC-MS]	[25-27]	RB-2	Students will learn the application of GC-MS
10	IV [Capillary electrophoresis]	[28-30]	RB-3	Students will learn about Capillary electrophoresis
11	IV [CE-MS]	[31-33]	RB-3	Students will learn about Capillary electrophoresis-Mass spectrometry
12	V [Supercritical Fluid Chromatography]	[34-36]	RB-4	Students will learn about Supercritical Fluid Chromatography
13	V [Supercritical Fluid Chromatography] VI [X-ray diffraction]	[37-39]	RB-4	Students will learn about Supercritical Fluid Chromatography and X-ray diffraction
14	VI [X-ray diffraction]	[40-42]	Journal Article 1	Students will learn about X-ray diffraction



7. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	30	90 min.	As per timetable	Close Book
Comprehensive Examination	30	180 min.	As per timetable	Close Book
Continuous assessment				
Assignment [I & II]	20	2 weeks	TBA	Open Book
Presentation [I & II]	20	10 min	TBA	Open Book

Criterion for NC: *Intentional Absence for any of the above evaluation components will attract award of NC*

8. Chamber Consultation Hour: To be announced in class.

9. Make-up Policy: Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. IN NO CASE THE MAKE-UP APPLICATION BE SLIPPED INSIDE THE CHAMBER OF THE INSTRUCTOR-IN-CHARGE. However, the decision of the Instructor-in-Charge in the above matter will be final.

Instructor-in-charge

Course No. PHA G534